Linear Algebra — Lecture 1

Systems of Linear Equations, Row Reduction, Vector Equations and Matrix Equations

Friedberg–Insel–Spence, *Linear Algebra* 4e, §1.4, §3.3–§3.4
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.1–§1.4

Topics. Systems of linear equations, row reduction and echelon forms, vector equations and matrix equations.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping for this lecture:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Systems of linear equations	§1.1 (p. 2)	§1.4 (pp. 24–28), §3.3 (p. 168)
Row reduction, echelon form	§1.2 (p. 12)	§3.4 (pp. 182–198)
Vector equations, linear combinations	§1.3 (p. 24)	§1.4 (pp. 24–31)
Matrix equation $A\boldsymbol{x} = \boldsymbol{b}$	§1.4 (p. 34)	§3.3 (pp. 168–181)

Two questions that drive the whole lecture: (1) *Existence*: is the system consistent? (2) *Uniqueness*: if consistent, is the solution unique?

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. For Lay, PDF page = printed page +17; for Friedberg, PDF page = printed page +15.

1 Systems of Linear Equations

Definition 1.1 (Linear equation and linear system; Lay, §1.1, p. 2)**.** A **linear equation** in the variables $x_1, \dots, x_n$ is one that can be written as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, \dots, a_n, b$ are real (or complex) constants. A collection of such equations is a **system of linear equations** (or a *linear system*).

A list $(s_1, \dots, s_n)$ is a **solution** if it satisfies *every* equation simultaneously; the set of all solutions is the **solution set**. Two systems are **equivalent** if they have the same solution set. A system is **consistent** if it has at least one solution, and **inconsistent** otherwise.

Proposition 1.1 (Only three possibilities; Lay, Theorem 2, §1.2, p. 21)**.** *A system of linear equations has either*

(i) no solution,
(ii) exactly one solution, or
(iii) infinitely many solutions.

Definition 1.2 (Matrices of a system; Lay, §1.1, p. 4)**.** For the system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \quad\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

the **coefficient matrix** and the **augmented matrix** are

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \qquad \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right].$$

An $m \times n$ matrix has m rows and n columns; the **size** is always *rows first.*

Definition 1.3 (Elementary row operations; Lay, §1.1, p. 6)**.**

1. **Replacement:** add to one row a multiple of another row;
2. **Interchange:** swap two rows;
3. **Scaling:** multiply all entries of a row by a nonzero constant.

Remark 1.1 (All three elementary row operations are reversible; Lay, §1.1, p. 6)**.** Each of the three operations is *reversible* and therefore produces an **equivalent** system: the solution set is unchanged.

Remark 1.2 (Why row operations are legitimate; Friedberg, Theorem 3.13, §3.4, p. 182)**.** Lay merely asserts that the three operations preserve the solution set; Friedberg turns this into a theorem (Chapter 3). Each elementary row operation amounts to **left multiplication by an elementary matrix** (§3.1, p. 148), which gives

Theorem 3.13 (p. 182). Let $A\boldsymbol{x} = \boldsymbol{b}$ be a system of m equations in n unknowns and let C be an invertible $m \times m$ matrix. Then the system $(CA)\boldsymbol{x} = C\boldsymbol{b}$ is **equivalent** to $A\boldsymbol{x} = \boldsymbol{b}$ (same solution set).

2 Row Reduction and Echelon Forms

Definition 2.1 (Echelon form; Lay, §1.2, p. 13)**.** A matrix is in **echelon form** (row echelon form) if

1. all nonzero rows are above any rows of all zeros;
2. each leading entry of a row is in a column to the right of the leading entry of the row above it;
3. all entries in a column below a leading entry are zero.

It is in **reduced echelon form** if, in addition,

4. the leading entry in each nonzero row is 1;
5. each leading 1 is the only nonzero entry in its column.

Definition 2.2 (Pivot; Lay, §1.2, p. 14)**.** A **pivot position** in A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A **pivot column** is a column of A containing a pivot position. Variables corresponding to pivot columns are **basic variables**; the others are **free variables**.

Algorithm 2.1 (The row reduction algorithm; Lay, §1.2, p. 15)**.**

Input: an $m \times n$ matrix A.
Output: the reduced echelon form R of A, still $m \times n$.

1. *scan from the left for the first column containing a nonzero entry; the top of that column is the first pivot position;*
2. *interchange rows so that the pivot row stands at the top, then use replacements to turn every entry below the pivot into* 0*;*
3. *cover the pivot row and every row above it, and repeat steps 1 and 2 on the remaining submatrix until no nonzero row is left;*
4. *starting from the rightmost pivot, use replacements to turn the entries above it into* 0*;*
5. *divide each pivot row by its pivot so that the pivot becomes* 1*.*

Theorem 2.1 (Uniqueness of the reduced echelon form; Lay, §1.2, p. 13)**.** *Every matrix is row equivalent to one and only one reduced echelon matrix.*

Proof. First, existence. Run Algorithm 2.1: in the forward phase each step moves the pivot column strictly right, so after at most $\min\{m, n\}$ steps the matrix is in echelon form; in the backward phase the entries above each pivot are cleared from the right and the pivots are scaled to 1, again in finitely many steps. The resulting matrix satisfies the three requirements of a reduced echelon form.

Second, uniqueness. Let R, S be row equivalent and in reduced echelon form; the three steps below match them row by row.

(i) The row space does not change. Write $\mathrm{Row}(A)$ for the subspace spanned by the rows of A. Interchanging rows and scaling a row only reorder or rescale the generators, and a replacement puts in a combination of two rows; none of the three changes the span, so $\mathrm{Row}(R) = \mathrm{Row}(S)$.

(ii) The pivot columns are determined by the row space. Let the nonzero rows of R be $\boldsymbol{r}_1, \dots, \boldsymbol{r}_k$ and its pivot columns $p_1 < \cdots < p_k$, and take $\boldsymbol{x} = \sum_{i=1}^{k} c_i \boldsymbol{r}_i \neq \boldsymbol{0}$ with i the smallest index such that $c_i \neq 0$. For $j < i$ the coefficient c_j is 0; for $j \geq i$ the row $\boldsymbol{r}_j$ is zero to the left of p_i (for $j = i$ because p_i is its first nonzero position, for $j > i$ because $p_j > p_i$). Hence $\boldsymbol{x}$ is zero to the left of p_i. In column p_i only $\boldsymbol{r}_i$ has entry 1, so the entry of $\boldsymbol{x}$ there is $c_i \neq 0$. Thus p_i is the first nonzero position of $\boldsymbol{x}$; conversely each p_i is the first nonzero position of $\boldsymbol{r}_i$, so

$$\{p_1, \dots, p_k\} = \{\text{first nonzero positions of the nonzero vectors in the row space}\}.$$

The right-hand side depends on the row space alone, so R and S have the same pivot columns and the same number k of nonzero rows.

(iii) Row by row. Write the nonzero rows of S as $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$. Fix i and put $\boldsymbol{d} = \boldsymbol{r}_i - \boldsymbol{s}_i$. The rows $\boldsymbol{r}_i$ and $\boldsymbol{s}_i$ both have entry δ_{ij} in column p_j, so $\boldsymbol{d}$ is zero in all those columns. On the other hand $\boldsymbol{d}$ lies in the common row space; write $\boldsymbol{d} = \sum_j c_j \boldsymbol{r}_j$. Its entry in column p_j is c_j, hence every $c_j = 0$, $\boldsymbol{d} = \boldsymbol{0}$ and $\boldsymbol{r}_i = \boldsymbol{s}_i$. The remaining rows are zero on both sides, so $R = S$. □

Remark 2.1 (Echelon forms are not unique; the reduced one is; Lay, §1.2, p. 13)**.** A matrix can have several echelon forms (Theorem 2.1): changing the order of elimination leaves different nonzero entries, and even scaling one row by 2 keeps the matrix in echelon form. Once the pivots are normalized to 1 and the entries above them are cleared, the result is unique.

This uniqueness is what makes the word "pivot" meaningful. Definition 2.2 calls a pivot position a location that matches a leading 1 in the reduced echelon form; without uniqueness the pivot columns, the basic variables and the free variables would have no definite content. Reading a solution set off the reduced form, and counting pivots to get the rank, rest on the same fact.

Theorem 2.2 (Existence and Uniqueness Theorem; Lay, Theorem 2, §1.2, p. 21)**.** *A linear system is consistent if and only if the rightmost column of the augmented matrix is* ***not*** *a pivot column, equivalently, if and only if an echelon form of the augmented matrix has no row of the form*

$$\left[\begin{array}{ccc|c} 0 & \cdots & 0 & b \end{array}\right] \qquad (b \neq 0).$$

If a system is consistent, then its solution set contains either

(i) a unique solution, when there are no free variables, or
(ii) infinitely many solutions, when there is at least one free variable.

Example 2.1 (A complete row reduction; Lay, §1.1, Examples 1–2, p. 4–7)**.** Solve

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ -4x_1 + 5x_2 + 9x_3 = -9 \end{cases}$$

Solution.

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Every column is a pivot column, so there are no free variables. The solution is therefore unique: $(x_1, x_2, x_3) = (29, 16, 3)$. □

Approach. Do not set out to "solve" the system: reduce the augmented matrix to reduced echelon form instead, handling one pivot column at a time and doing only two things per column: scale the pivot to 1, then clear both above and below it.

Remark 2.2 (Parametric description; Lay, §1.5, p. 44–45)**.** When free variables occur, solve each pivot-variable equation for its basic variable in terms of the free ones. Writing $x_3 = t$ etc. gives the general solution in parametric (vector) form; this is the standard way to describe an infinite solution set.

3 Vector Equations

Vectors in $\mathbb{R}^n$ are $n \times 1$ column matrices. For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$, addition and scalar multiplication are defined componentwise. Geometrically, $\boldsymbol{u} + \boldsymbol{v}$ is given by the **parallelogram rule**, and $c\boldsymbol{v}$ is a scaling (reversing direction when $c < 0$).

Definition 3.1 (Linear combination; Lay, §1.3, p. 27)**.** Given $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p \in \mathbb{R}^n$ and scalars $c_1, \ldots, c_p$, the vector

$$\boldsymbol{y} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_p\boldsymbol{v}_p$$

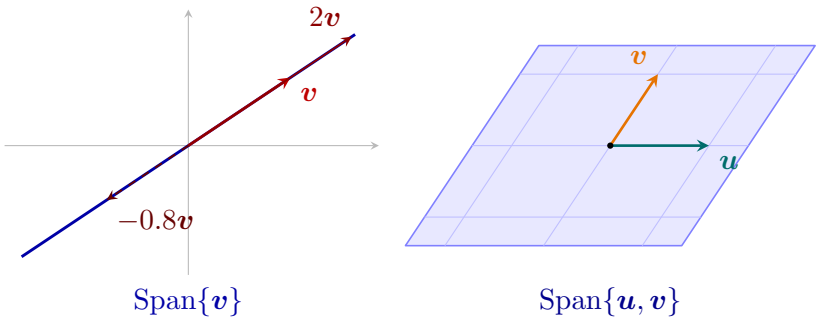
is a **linear combination** of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ with **weights** $c_1, \ldots, c_p$.

Definition 3.2 (Span; Lay, §1.3, p. 30)**.** If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p \in \mathbb{R}^n$, the set of all linear combinations of them is denoted $\mathrm{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ and is called the subset of $\mathbb{R}^n$ **spanned** (or generated) by $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$.

Proposition 3.1 (The key translation; Lay, §1.3, p. 30–31)**.** *For $\boldsymbol{b} \in \mathbb{R}^n$, the following are equivalent:*

(i) $\boldsymbol{b} \in \mathrm{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$;
(ii) the vector equation $x_1\boldsymbol{v}_1 + \cdots + x_p\boldsymbol{v}_p = \boldsymbol{b}$ has a solution;
(iii) the linear system with augmented matrix $\left[\begin{array}{ccc|c} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_p & \boldsymbol{b} \end{array}\right]$ is consistent.

Remark 3.1 (Geometry; Lay, §1.3, p. 28–29)**.** Span$\{\boldsymbol{v}\}$ ($\boldsymbol{v} \neq \mathbf{0}$) is a line through the origin; Span$\{\boldsymbol{u}, \boldsymbol{v}\}$ with $\boldsymbol{u}, \boldsymbol{v}$ non-proportional is a plane through the origin. Every span contains $\mathbf{0}$ (take all weights zero).



(a) Span$\{\boldsymbol{v}\}$ is a line through the origin; (b) when $\boldsymbol{u}, \boldsymbol{v}$ are not proportional, Span$\{\boldsymbol{u}, \boldsymbol{v}\}$ is a plane through the origin.

Remark 3.2 (The abstract version: Friedberg §1.4, pp. 24–31)**.** Lay works inside $\mathbb{R}^n$; Friedberg defines the same notions for a **vector space** V **over an arbitrary field** $\mathbb{F}$:

- For nonempty $S \subseteq V$, a vector v is a **linear combination** of vectors of S if $v = a_1 u_1 + \cdots + a_n u_n$ for some $u_1, \ldots, u_n \in S$ and $a_1, \ldots, a_n \in \mathbb{F}$; the a_i are the **coefficients**.
- The **span** Span(S) is the set of all linear combinations of vectors in S, with the convention Span$(\emptyset) = \{\mathbf{0}\}$.
- **Theorem 1.5** (p. 30): Span(S) is a **subspace** of V, and every subspace containing S also contains Span(S), i.e. Span(S) is the **smallest subspace containing** S.
- If Span$(S) = V$, then S **generates** (or spans) V.

Worth comparing with the book's examples (pp. 31–32): $(1,1,0), (1,0,1), (0,1,1)$ generate $\mathbb{R}^3$; three polynomials generate $P_2(\mathbb{R})$; four matrices generate $M_{2\times 2}(\mathbb{R})$, whereas three matrices all having equal diagonal entries do *not* generate $M_{2\times 2}(\mathbb{R})$.

4 The Matrix Equation $A\boldsymbol{x} = \boldsymbol{b}$

Definition 4.1 (The product $A\boldsymbol{x}$; Lay, §1.4, p. 35)**.** If A is an $m \times n$ matrix with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$, and if $\boldsymbol{x} \in \mathbb{R}^n$, then

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n.$$

Remark 4.1 (Read $A\boldsymbol{x}$ as a combination of the columns; Lay, §1.4, p. 35)**.** Read every occurrence of $A\boldsymbol{x}$ as "a linear combination of the columns of A." This single viewpoint is the one that survives into abstract vector spaces later in the course.

Theorem 4.1 (Three equivalent formulations; Lay, Theorem 3, §1.4, p. 36)**.** *Let A be $m \times n$ with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ and let $\boldsymbol{b} \in \mathbb{R}^m$. Then the matrix equation $A\boldsymbol{x} = \boldsymbol{b}$, the vector equation $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}$, and the linear system with augmented matrix $\left[\begin{array}{ccc|c} \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n & \boldsymbol{b} \end{array}\right]$ all have the **same solution set**.*

Theorem 4.2 (Four equivalent statements; Lay, Theorem 4, §1.4, p. 37)**.** *Let A be an $m \times n$ matrix. The following are logically equivalent: for a given A, either all are true or all are false.*

(a) For each $\boldsymbol{b} \in \mathbb{R}^m$, the equation $A\boldsymbol{x} = \boldsymbol{b}$ has a solution.
(b) Each $\boldsymbol{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
(c) The columns of A span $\mathbb{R}^m$.
*(d) A has a pivot position in **every row**.*

Theorem 4.3 (Consistency and structure of the solution set; Lay, Theorem 6, §1.5, p. 46; Friedberg, Theorem 3.9, §3.3, p. 172)**.** *Let A be $m \times n$ and $\boldsymbol{b} \in \mathbb{R}^m$. Then:*

(i) $A\boldsymbol{x} = \boldsymbol{b}$ is consistent iff $\boldsymbol{b}$ is a linear combination of the columns of A;
(ii) if $A\boldsymbol{x} = \boldsymbol{b}$ is consistent, its solution set is the translate of the solution set of $A\boldsymbol{x} = \boldsymbol{0}$:

$$\{\boldsymbol{x}\} = \boldsymbol{p} + \{\boldsymbol{x}_h : A\boldsymbol{x}_h = \boldsymbol{0}\},$$

where $\boldsymbol{p}$ is any particular solution.

Definition 4.2 (Identity matrix; Lay, §1.4, p. 38)**.** The **identity matrix** I_n is the $n \times n$ matrix with 1's on the diagonal and 0's elsewhere; it satisfies $I_n\boldsymbol{x} = \boldsymbol{x}$ for every $\boldsymbol{x} \in \mathbb{R}^n$.

5 A Worked Example

Example 5.1 (Consistency, pivots and spanning; This example was written for this lecture; the methods are Lay §1.1–§1.4)**.** Let

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 1 & 4 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

and write the three columns of A as $\boldsymbol{v}_1 = (1, 0, 1)^\top$, $\boldsymbol{v}_2 = (3, 1, 4)^\top$, $\boldsymbol{v}_3 = (2, -1, 1)^\top$. Find the condition for $A\boldsymbol{x} = \boldsymbol{b}$ to be consistent, decide whether the solution is then unique, and decide whether the three columns span $\mathbb{R}^3$.

Solution. (1) Find the condition for $A\boldsymbol{x} = \boldsymbol{b}$ to be consistent. Row reduce the augmented matrix to reduced echelon form (Lay §1.2, p. 12):

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & b_1 \\ 0 & 1 & -1 & b_2 \\ 1 & 4 & 1 & b_3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & b_1 \\ 0 & 1 & -1 & b_2 \\ 0 & 1 & -1 & b_3 - b_1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & b_1 - 3b_2 \\ 0 & 1 & -1 & b_2 \\ 0 & 0 & 0 & b_3 - b_1 - b_2 \end{array}\right].$$

The last row reads $0 = b_3 - b_1 - b_2$, so the system is **consistent exactly when** $b_3 = b_1 + b_2$; the case $b_3 \neq b_1 + b_2$ is the "no solution" branch.

(2) Decide whether the solution is unique when the system is consistent. The pivot columns of the reduced form are columns 1 and 2, while the number of unknowns is $n = 3$, so x_3 is free:

$$x_1 = b_1 - 3b_2 - 5x_3, \qquad x_2 = b_2 + x_3 \qquad (x_3 \text{ arbitrary}),$$

hence there are **infinitely many solutions**. That is the trichotomy of Lay §1.1, p. 2 in action.

(3) Decide whether the three columns span $\mathbb{R}^3$. Column 3 is not a pivot column, so $\boldsymbol{v}_3$ is a linear combination of the first two; the reduced form reads it off:

$$\boldsymbol{v}_3 = 5\boldsymbol{v}_1 - \boldsymbol{v}_2.$$

so

$$\operatorname{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\} = \operatorname{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} = \{(b_1, b_2, b_3)^\top : b_3 = b_1 + b_2\},$$

a **plane** through the origin in $\mathbb{R}^3$; the three vectors therefore do **not** span $\mathbb{R}^3$.

The same echelon form answers three questions – whether the augmented column creates a contradictory row decides **consistency**, the number of pivot columns decides the **degrees of freedom**, and which columns are *not* pivots decides **which dimension the span is missing**. Friedberg writes the first of these as an equation between ranks (Theorem 3.11, §3.3, p. 174). □

Approach. When the question is "for which $\boldsymbol{b}$ is the system consistent", do not first guess a form for $\boldsymbol{b}$, reduce the coefficient matrix to reduced echelon form instead. That one echelon form answers all three questions at once: whether the augmented column creates a contradictory row (consistency), how many columns are not pivots (degrees of freedom), and which dimension the columns' span is missing.

6 Checklist and Exercises

By the end of this lecture you should be able to:

1. Write the augmented matrix of a system and reduce it to reduced echelon form without errors.
2. Read off from an echelon form whether the system has no solution, one solution, or infinitely many, and produce the general (parametric) solution.
3. Explain the correspondence between pivot columns, basic variables and free variables.
4. Translate "is $\boldsymbol{b} \in \text{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$?" into a question about consistency of an augmented matrix.
5. State the definition of $A\boldsymbol{x}$ and explain Theorem 4.1.
6. Use condition (d) of Theorem 4.2 to decide whether the columns of A span $\mathbb{R}^m$.

Exercises (printed pages). §1.1 p. 10 · §1.2 p. 21 · §1.3 p. 32 · §1.4 p. 40. Each section also ends with *Practice Problems* before the exercise set; do those first.

How to use the two texts together:

- **Lay for computation, Friedberg for proof.** The same result carries different numbers in the two books; these notes give both.
- Friedberg works over an **arbitrary field** $\mathbb{F}$, not just $\mathbb{R}$. When a problem involves $\mathbb{C}$ or $\mathbb{Z}_p$, only Friedberg's statements apply; Lay's conclusions do not transfer automatically.
- The core chain of this lecture: *augmented matrix* $\rightarrow$ *echelon form* $\rightarrow$ *pivots decide existence and uniqueness* $\rightarrow$ *pivot and free columns describe* Span.